Skills Summary

Finding Unit Rates with Whole-Number Quantities

Use this reference while completing your worksheet or when studying.

Skill 1 — Finding a unit rate by dividing

Rule: A *unit rate* tells you how much of one quantity goes with **exactly one** of the other.

$$\text{unit rate} = \frac{\text{total of the first quantity}}{\text{number of the second quantity}}$$

The quantity named after the word *per* is always the divisor: "dollars per bottle" means dollars $\div$ bottles.

Example: A 6-pack of juice costs 18 dollars. Find the cost per bottle.
Step 1. "Per bottle" names bottles as the divisor, so I divide the dollars by the number of bottles rather than the other way round.
Step 2. $18 \div 6 = 3$, and the units come along: dollars per bottle. $\Rightarrow$ **3** dollars per bottle

Try it: A printer prints 40 pages in 8 minutes. How many pages per minute?

check: **5** pages per minute

Skill 2 — Reading a unit rate out of a ratio table

Rule: Every column of a ratio table holds the same rate. To reach the "per one" column, divide **both** rows by the same number; to leave it, multiply both rows by the same number.

Example: A car covers 208 miles in 4 hours.

Hours	4	1
Miles	208	?

Step 1. The table already pairs the two quantities, so I just scale the hours column down to 1 and do the same thing to the miles.
Step 2. $4 \div 4 = 1$, so $208 \div 4 = 52$. $\Rightarrow$ **52** miles per hour

Try it: 7 identical boxes hold 91 crayons. How many crayons per box?

check: **13** crayons per box

Skill 3 — Comparing two rates fairly

Rule: Two totals can only be compared when they buy the same amount. Bring *both* offers down to one unit, then compare the unit rates. Subtracting the two unit rates says by how much, and a difference of 0 means the two rates are equal.

Example: 3 lb for 12 dollars, or 5 lb for 15 dollars? By how much per pound?
Step 1. The two offers weigh different amounts, so the totals are not comparable — I reduce each to dollars per pound first.
Step 2. $12 \div 3 = 4$ and $15 \div 5 = 3$; the difference is $4 - 3$. $\Rightarrow$ **1** dollar more per pound at the first store

Try it: 4 lb for 32 dollars, or 5 lb for 30 dollars? Subtract the second rate from the first.

check: **2** dollars more per pound

Skill 4 — Using a unit rate to scale up

Rule: Once you have the rate for one, multiply it by however many you want. The same work written as a proportion is $\frac{p}{q} = \frac{x}{n}$, which cross-multiplies to $qx = pn$.

Example: 9 notebooks cost 45 dollars. What do 13 notebooks cost?
Step 1. The two orders differ only in how many notebooks, so one notebook is the bridge between them.
Step 2. $45 \div 9 = 5$ dollars each, then $5 \times 13 = 65$; as a proportion, $9x = 45 \cdot 13$ gives the same x. $\Rightarrow$ **65** dollars

Try it: A printer prints 8 pages in 2 minutes. How many minutes for 52 pages?

check: **13** minutes